



Gorilla Glass Victus

Corning's latest iteration of glass for electronics can withstand drops from twice the height as Gorilla Glass 6. <https://dgit.in/aug20-59>



Dell gaming laptops

Dell has launched a new lineup of gaming laptops, including the Ryzen powered G5 15 SE. <https://dgit.in/aug20-60>

AMD Ryzen 9 3900XT

Refined silicon incoming

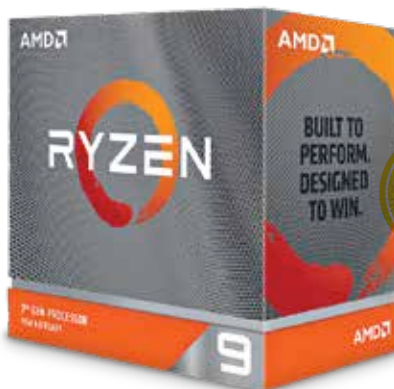
AMD announced the new XT lineup a couple of weeks ago with increased boost clock speeds and improvements in the silicon that were geared towards helping the processors maintain higher clock residency. Three new processors would be added to the XT lineup in the form of the AMD Ryzen 5 3600XT, Ryzen 7 3800XT and Ryzen 9 3900XT. The best part is that AMD stated that these new XT parts would be priced the same as their X contemporaries were at launch.

AMD 3rd Gen RYZEN 9 / 7 / 5 Matisse XT Refresh Lineup and Specifications

AMD new XT lineup brings incremental changes in the specifications over their X variants. There really isn't a need to bump up the prices of the processors for such a miniscule improvement. The competition could really take a page out of AMD's book in this regard. The Ryzen 5 3600XT and Ryzen 9 3900XT bring single bin increments in the Boost Clock whereas the Ryzen 7 3800XT shows twice as much improvement. The base clock speeds are the same between the X and XT variants.

International pricing for the SKUs has been kept the same. The Ryzen 5 3600X and 3600XT, have a launch price of USD 249 but we're more concerned with the Indian pricing. The Ryzen 5 3600X was priced at INR 20,874 at launch and the new Ryzen 5 3600XT is priced at INR 23,588. Considering that these are imports and dependent on USD-INR rates, it seems that the two are, in fact, priced the same when you factor in the USD-INR rates.

The same goes for the AMD Ryzen 7 3800XT and Ryzen 9 3900XT. However, there's one key difference here. The X variants, i.e. the 3800X and 3900X, shipped with AMD's Wraith



Price
₹46,000



PERFORMANCE.....90
VALUE FOR MONEY.....75

Prism coolers whereas the XT variants don't. The Amazon.

com pricing for the AMD Wraith Prism cooler is USD 41.99, so there is a marked difference in the value proposition of the 3800XT and 3900XT processors. What should also be noted is that folks going for high-end desktop processors often opt for better third-party coolers so the exchange of losing out on a cooler for slightly faster clock speed is something that the consumer might want to consider while deciding between the X and XT variants.

AMD RYZEN 9 3900XT Performance

The AMD Ryzen 9 3900XT is an incremental upgrade over the Ryzen 9 3900X since only the boost clock speed has increased. However, there have been several improvements to the silicon within which allows the Ryzen 9 3900XT to make better use of this slight improvement. Let's see how that pans out in the benchmarks.

We tested the unit on the following configuration – CPU: AMD Ryzen 9 3900XT, Motherboard: ASUS ROG STRIX X570-E Gaming, SSD: Samsung 970 PRO 1TB, HDD: Seagate FireCuda 2 TB ST2000DX002, RAM: Corsair Dominator PRO 16 GB (2x 8GB) @3200 MHz, Cooler: Corsair H115i Pro, and PSU: Corsair HX1050.

As always, we begin with Cinebench 20. In the single-threaded benchmark, The AMD Ryzen 9 3900XT ends up performing about 3.5 per cent better than the Ryzen 3900X and retains a lead over the Intel Core i9-9900K. When compared to the Intel Core i9-10900K, the two are neck and neck in single threaded performance but the tables turn in the multi-threaded benchmark as the AMD Ryzen 9 3900XT races far ahead.

Our Handbrake run basically consists of transcoding a 4K video clip from one CODEC to another without changing the resolution. The 3900XT shows a marked reduction in encode times over the Ryzen 9 3900X.

Blender is a fairly well known open-source rendering software. In this benchmark, we're looking at the time taken for a run to finish rendering. So the lower the score in this test, better the CPU. The AMD Ryzen 9 3900X ends the run a minute earlier than the Ryzen 9 3900X which is again, an incremental improvement.

In our gaming benchmarks, we see an average of 1-2 FPS improvement with the Ryzen 9 3900XT over the Ryzen 9 3900X when running at 1080p.

Bottomline

For folks wanting to build a new PC, the AMD Ryzen 9 3900XT is a fairly obvious choice over the vanilla Ryzen 9 3900X owing to the slight bump in boost clock speeds. Unlike the 3900X, the 3900XT does not come with a Wraith Prism cooler, so that's something PC builders might want to factor into their buying decisions. Aside from that, the Ryzen 9 3900XT is a great processor for PCs geared towards gaming, content creation, 3D modelling and multi-tasking, in general.

—Mithun Mohandas

SPECIFICATIONS

CORES: 12 | THREADS: 24 | BASE CLOCK: 3.8 GHz | BOOST CLOCK: 4.7 GHz | PROCESS NODE: 7nm | L3 CACHE: 64 MB | L2 CACHE: 512 KB per core | TDP: 105 W

CONTACT

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